A dot plot has most of its dots on the left and a long tail to the right.

How is its shape described?

A. Uniform

B. Skewed right

C. Skewed left

D. Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

1601

QUESTION ID

S52-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Is the distribution in the dot plot symmetric?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

1602

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q2/5

A dot plot has a tall stack of dots in the middle and short, equal stacks on each side.

What is the shape of the distribution?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

1603

QUESTION ID

S52-B2-Q3

TYPE

Short answer

CORRECT ANSWER

Symmetric (mound-shaped)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q3/5

True or false?

A distribution that looks the same on both sides of its center is symmetric.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

1604

QUESTION ID

S52-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q4/5

True or false?

A dot plot with all of its dots stacked above one value is skewed left.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

1605

QUESTION ID

S52-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 7.SP.B.3

FORWARD 1

Compare groups using relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The table shows how students get to school.

What percent of sixth graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

1607

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows how students get to school.

What percent of seventh graders walk?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

1608

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

25%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows how students get to school.

Is there evidence that grade and walking are related?

	Walks	Rides	Total
Sixth grade	24	36	60
Seventh grade	15	45	60
Total	39	81	120

A. No, because each grade has 60 students.

B. No, because more students ride than walk in both grades.

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

D. Yes, because 24 is greater than 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

1609

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes, because 40% of sixth graders walk but only 25% of seventh graders do.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

Relative frequencies make it possible to compare groups of different sizes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

1610

QUESTION ID

S52-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If 40% of sixth graders and 40% of seventh graders walk, there is a strong relationship between grade and walking.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

1611

QUESTION ID

S52-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 52 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 52 • 7.SP.B.3

FORWARD 2

Compare the center and spread of two data sets • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Data set A has a mean of 20 and a standard deviation of 2. Data set B has a mean of 20 and a standard deviation of 6.

Which data set is more spread out?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

1613

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Data set B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q1/5

Data set A has a median of 45 and an IQR of 10. Data set B has a median of 52 and an IQR of 10.

Compare the centers and the spreads.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

1614

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

B's median is 7 greater; the spreads (IQRs) are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q2/5

Which measure of spread is best for a skewed data set with outliers?

A. Interquartile range

B. Mean

C. Range

D. Standard deviation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

1615

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q3/5

True or false?

Standard deviation measures how spread out the data are around the mean.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

1616

QUESTION ID

S52-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q4/5

True or false?

Two data sets with the same mean must have the same standard deviation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

1617

QUESTION ID

S52-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 52 • FORWARD 2 • Q5/5

SET 53 • 7.SP.B.4 • STATISTICS & PROBABILITY

Compare two populations using measures of center and variability from random samples

MAIN	Grade 7 main question	**7.SP.B.4**
BACKWARD 1	Find the interquartile range	**6.SP.B.5.c**
BACKWARD 2	Read a box plot	**6.SP.B.4**
FORWARD 1	Identify clusters and outliers in a scatter plot — nearest related (n…	**8.SP.A.1**
FORWARD 2	Interpret differences in center and spread, including the effect of…	**HSS-ID.A.3**

SET 53 • 7.SP.B.4

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.B.4

Random samples of quiz scores: Class A has a median of 78 and an IQR of 10. Class B has a median of 70 and an IQR of 10.

Which class tends to score higher, and by how much (median)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

1620

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

Class A, by 8 points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q1/5

A random sample of 7th graders has a mean of 6.5 hours of screen time and a MAD of 1.2. A random sample of 8th graders has a mean of 7.7 hours and a MAD of 1.2.

Write the difference in means as a multiple of the MAD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

1621

QUESTION ID

S53-M-Q2

TYPE

Short answer

CORRECT ANSWER

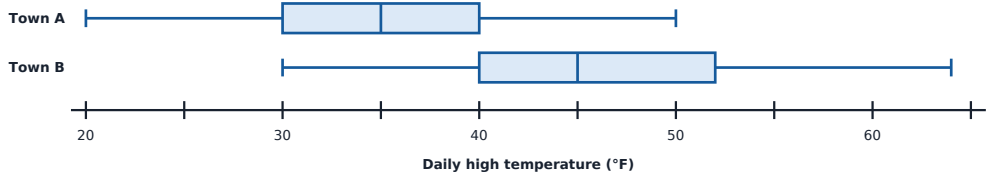
1 MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q2/5

The box plots show random samples of daily high temperatures in two towns.

Which statement is true?



A. The two towns have the same range.

B. The IQRs differ by 10 degrees.

C. The median for Town B is 10 degrees greater than the median for Town A.

D. Town A has the greater median.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

1622

QUESTION ID

S53-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The median for Town B is 10 degrees greater than the median for Town A.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-M-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q3/5

True or false?

Random samples are a good way to compare two populations.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

1623

QUESTION ID

S53-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q4/5

True or false?

Sample A has a greater mean than Sample B. So every value in Sample A is greater than every value in Sample B.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

1624

QUESTION ID

S53-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 53 • MAIN • Q5/5

SET 53 • 7.SP.B.4

BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

A data set has Q1 = 12 and Q3 = 20.

What is the IQR?

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

1626

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q1/5

Find the IQR.

2, 4, 6, 8, 10, 12, 14, 16

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

1627

QUESTION ID

S53-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q2/5

A box plot has Q1 = 30 and Q3 = 55.

What is the IQR?

A. 30

B. 85

C. 25

D. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

1628

QUESTION ID

S53-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q3/5

True or false?

The IQR is the difference between Q3 and Q1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

1629

QUESTION ID

S53-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q4/5

True or false?

A data set with Q1 = 7 and Q3 = 15 has an IQR of 22.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

1630

QUESTION ID

S53-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The IQR is 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 53 • BACKWARD 1 • Q5/5

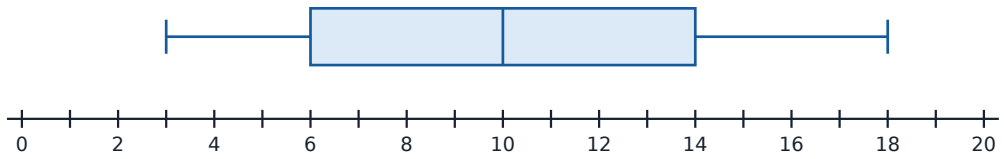
SET 53 • 7.SP.B.4

BACKWARD 2

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

ANSWER KEY Reduced copy of the question page, with its answer

What is the median of the data in the box plot?



Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

1632

QUESTION ID

S53-B2-Q1

TYPE

Short answer

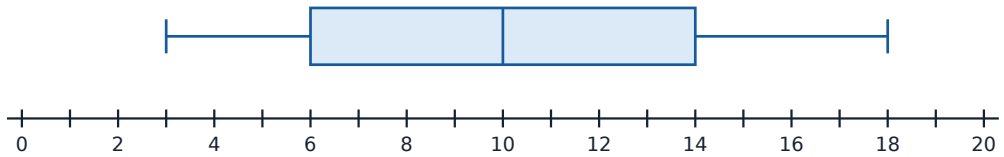
CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q1/5

What is the first quartile (Q1) of the data in the box plot?



Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

1633

QUESTION ID

S53-B2-Q2

TYPE

Short answer

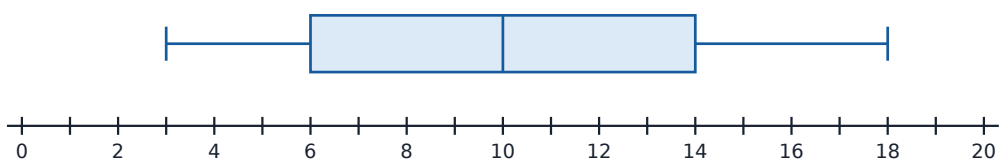
CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q2/5

What is the maximum of the data in the box plot?



A. 14

B. 18

C. 3

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

1634

QUESTION ID

S53-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q3/5

True or false?

The box of a box plot shows the middle half of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

1635

QUESTION ID

S53-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

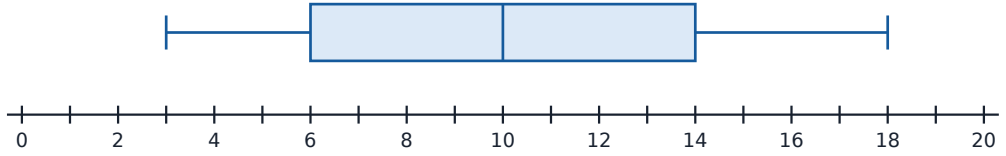
ID S53-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The minimum of the data in the box plot is 6.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

1636

QUESTION ID

S53-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The minimum is 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 7.SP.B.4

FORWARD 1

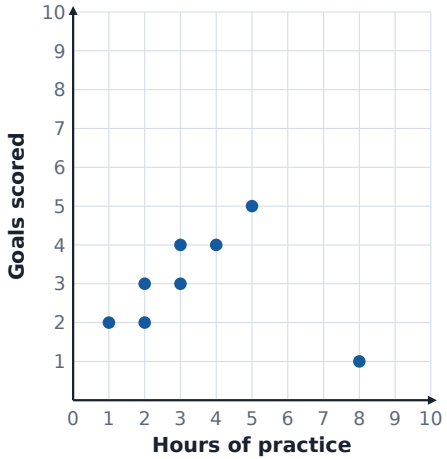
Identify clusters and outliers in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

Which point in the scatter plot is an outlier?

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

1638

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

(8, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q1/5 • NEAREST RELATED

A scatter plot has a group of points close together.

What is the group called?

A. A cluster

B. An outlier

C. An intercept

D. A slope

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

1639

QUESTION ID

S53-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

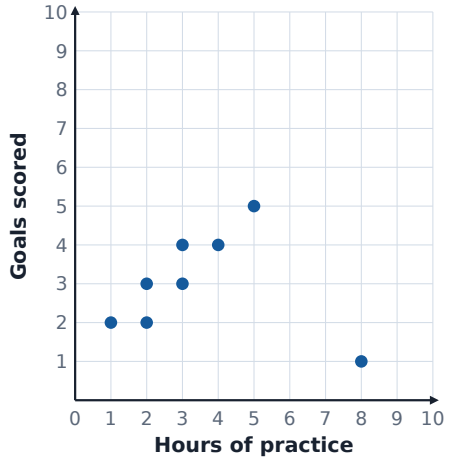
A. A cluster

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q2/5 • NEAREST RELATED

Leaving out the outlier, describe the association in the scatter plot.

Association: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

1640

QUESTION ID

S53-F1-Q3

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

An outlier is a point that is far from the overall pattern of the data.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

1641

QUESTION ID

S53-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

Every scatter plot has exactly one outlier.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

1642

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 53 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 53 • 7.SP.B.4

FORWARD 2

Interpret differences in center and spread, including the effect of outliers • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set is 4, 5, 5, 6, 7, 40.

Which better describes its center, the mean or the median? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

1644

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The median, because the outlier 40 pulls the mean up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q1/5

A data set is 4, 5, 5, 6, 7, 40.

What is the mean with the outlier, and what is the mean without it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

1645

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

About 11.2 with the outlier; 5.4 without it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q2/5

Which measure is affected most by an outlier?

A. Mean

B. IQR

C. Mode

D. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

1646

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q3/5

True or false?

The median is resistant to outliers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

1647

QUESTION ID

S53-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q4/5

True or false?

Removing a very large outlier makes the mean larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

1648

QUESTION ID

S53-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S53-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3 • Set 53 • FORWARD 2 • Q5/5

SET 54 • 7.SP.C.5 • STATISTICS & PROBABILITY

Understand probability as a number from 0 to 1 that shows how likely an event is

MAIN	Grade 7 main question	**7.SP.C.5**
BACKWARD 1	Compare two fractions	**4.NF.A.2**
BACKWARD 2	Place fractions on a number line from 0 to 1	**3.NF.A.2**
FORWARD 1	Find relative frequencies in a two-way table — nearest related (no…	**8.SP.A.4**
FORWARD 2	Describe events as subsets of a sample space	**HSS-CP.A.1**

SET 54 • 7.SP.C.5

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.5

Which probability describes an event that is likely to happen?

A. 0.1

B. 0.8

C. 0.5

D. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

1651

QUESTION ID

S54-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 5 red marbles and no other marbles.

What is the probability of drawing a red marble?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

1652

QUESTION ID

S54-M-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q2

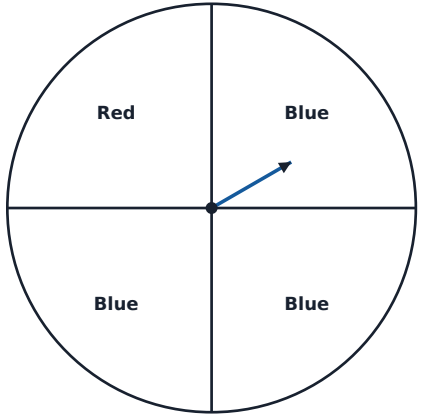
Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The spinner has 4 equal sections.

Is landing on Blue likely, unlikely, or neither likely nor unlikely?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

1653

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

Likely

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q3/5

True or false?

An event with a probability of $\frac{1}{2}$ is just as likely to happen as not to happen.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

1654

QUESTION ID

S54-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A probability of 1.3 means that an event is very likely.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

1655

QUESTION ID

S54-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Probabilities are from 0 to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 54 • MAIN • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 1

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is greater, $\frac{3}{8}$ or $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

1657

QUESTION ID

S54-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q1/5

Write >, <, or =.

$\frac{5}{10}$ ___ $\frac{1}{2}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

1658

QUESTION ID

S54-B1-Q2

TYPE

Short answer

CORRECT ANSWER

=

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q2/5

Which fraction is less than $\frac{1}{2}$?

A. $\frac{3}{4}$

B. $\frac{5}{8}$

C. $\frac{3}{8}$

D. $\frac{4}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

1659

QUESTION ID

S54-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q3/5

True or false?

$\frac{7}{12} > \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

1660

QUESTION ID

S54-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$\frac{2}{5} > \frac{3}{4}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

1661

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 7.SP.C.5

BACKWARD 2

Place fractions on a number line from 0 to 1 • Grade 3 • Number & Operations–Fractions • 3.NF.A.2

Plot $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

1663

QUESTION ID

S54-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A point at the third tick mark after 0 (3/4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q1/5

A number line from 0 to 1 is split into 4 equal parts.

What fraction is at the first tick mark after 0?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

1664

QUESTION ID

S54-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q2/5

Which fraction is halfway between 0 and 1 on a number line?

A. $\frac{1}{2}$

B. $\frac{1}{4}$

C. $\frac{2}{1}$

D. $\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

1665

QUESTION ID

S54-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q3/5

True or false?

$\frac{4}{4}$ is at the same point as 1 on a number line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

1666

QUESTION ID

S54-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3}$ is farther from 0 than $\frac{2}{3}$ is.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

1667

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.2 • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 7.SP.C.5

FORWARD 1

Find relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among Grade 9 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

1669

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 18/60 or 30% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among Grade 10 students, what is the relative frequency of playing music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

1670

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 12/40 or 30% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q2/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the relative frequency of all students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.3 **B.** 0.18 **C.** 0.12 **D.** 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

1671

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, students in both grades play music at the same rate.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

1672

QUESTION ID

S54-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the table, Grade 9 students are more likely than Grade 10 students to play music, because 18 > 12.

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

1673

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both rates are 30%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 54 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 54 • 7.SP.C.5

FORWARD 2

Describe events as subsets of a sample space • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1

A number cube is rolled.

List the outcomes in the event "an even number."

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

1675

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q1/5

A number cube is rolled.

List the outcomes in the complement of the event "a number greater than 4."

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

1676

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q2/5

A coin is flipped twice.

What is the sample space?

A. HH, HT, TH, TT

B. HH, TT

C. HH, HT, TT

D. H, T

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

1677

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. HH, HT, TH, TT

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S54-F2-Q3**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q3/5

True or false?

The probability of an event plus the probability of its complement is 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

1678

QUESTION ID

S54-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q4/5

True or false?

When a number cube is rolled, the events "even number" and "odd number" together include no outcomes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

1679

QUESTION ID
S54-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE Together they include all six outcomes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S54-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.1 • Set 54 • FORWARD 2 • Q5/5

SET 55 • 7.SP.C.6 • STATISTICS & PROBABILITY

Estimate a probability from relative frequency and make predictions

MAIN	Grade 7 main question	**7.SP.C.6**
BACKWARD 1	Use a rate to make a prediction	**6.RP.A.3.b**
BACKWARD 2	Compare decimals to thousandths	**5.NBT.A.3.b**
FORWARD 1	Use relative frequencies in a two-way table to make predictions…	**8.SP.A.4**
FORWARD 2	Find conditional probabilities from a two-way table	**HSS-CP.A.4**

SET 55 • 7.SP.C.6

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.6

ANSWER KEY Reduced copy of the question page, with its answer

A spinner landed on green 36 times in 120 spins.

Estimate the probability of landing on green.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

1682

QUESTION ID

S55-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

GRADING NOTE 36/120 or 3/10 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q1/5

A basketball player made 42 of her last 60 free throws.

Predict how many she will make in her next 150 free throws.

Free throws: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

1683

QUESTION ID

S55-M-Q2

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q2/5

A coin was flipped 200 times and landed heads 94 times.

What is the relative frequency of heads?

A. 0.94

B. 0.47

C. 0.53

D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

1684

QUESTION ID

S55-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q3/5

True or false?

A number cube is rolled 300 times. About 50 of the rolls should be a 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

1685

QUESTION ID

S55-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q4/5

True or false?

A spinner landed on red 9 out of 10 times. The probability of red must be exactly 0.9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

1686

QUESTION ID

S55-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Relative frequency only estimates the probability, and 10 spins is a small number of trials.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 55 • MAIN • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 1

Use a rate to make a prediction • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

A team scores 3 goals every 2 games.

At this rate, how many goals does it score in 10 games?

Goals: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

1688

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q1/5

4 out of every 10 apples in a crate are green.

At this rate, how many of 250 apples are green?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

1689

QUESTION ID

S55-B1-Q2

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q2/5

A machine fills 6 bottles every 4 seconds.

At this rate, how many bottles does it fill in 30 seconds?

A. 45

B. 20

C. 24

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

1690

QUESTION ID

S55-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q3/5

True or false?

At a rate of 2 out of every 5, there are 40 out of every 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

1691

QUESTION ID

S55-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q4/5

True or false?

At a rate of 3 out of every 8, there are 30 out of every 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

1692

QUESTION ID

S55-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE There are 24 out of every 64.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B1-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 7.SP.C.6

BACKWARD 2

Compare decimals to thousandths • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater, 0.47 or 0.5?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

1694

QUESTION ID

S55-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write >, <, or =.

0.305 ___ 0.35

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

1695

QUESTION ID

S55-B2-Q2

TYPE

Short answer

CORRECT ANSWER

<

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q2/5

Which decimal is the greatest?

A. 0.059

B. 0.62

C. 0.506

D. 0.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

1696

QUESTION ID

S55-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q3/5

True or false?

0.250 = 0.25

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

1697

QUESTION ID

S55-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q4/5

True or false?

0.09 > 0.1

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

1698

QUESTION ID

S55-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 7.SP.C.6

FORWARD 1

Use relative frequencies in a two-way table to make predictions • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

Among teens, what is the relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

1700

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.225

GRADING NOTE 9/40 or 22.5% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q1/5 • NEAREST RELATED

Use the adults row of the table.

Predict how many of 200 adults like rain.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

Adults: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

1701

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows the results of a survey.

Which group has the greater relative frequency of liking rain?

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. Adults

B. Teens

C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

1702

QUESTION ID

S55-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Adults

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of liking rain among all 100 people is 0.27.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

1703

QUESTION ID

S55-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of disliking rain among adults is 0.42.

	Likes rain	Dislikes rain	Total
Adults	18	42	60
Teens	9	31	40
Total	27	73	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

1704

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 42/60 = 0.7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 55 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 55 • 7.SP.C.6

FORWARD 2

Find conditional probabilities from a two-way table • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4

ANSWER KEY Reduced copy of the question page, with its answer

The table shows students' favorite pets. A student is chosen at random.

Find P(dog | Grade 9).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

1706

QUESTION ID

S55-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

GRADING NOTE 25/40 or 0.625 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows students' favorite pets. A student is chosen at random.

Find P(Grade 10 | cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

1707

QUESTION ID

S55-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/7

GRADING NOTE 20/35 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S55-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q2/5

The table shows students' favorite pets. A student is chosen at random.

Find P(cat).

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. $\frac{1}{2}$ **B.** $\frac{9}{16}$ **C.** $\frac{7}{16}$ **D.** $\frac{3}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

1708

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 7/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q3/5

True or false?

In the table, P(dog | Grade 10) = $\frac{1}{2}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

1709

QUESTION ID

S55-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

In the table, P(Grade 9 | dog) = $\frac{25}{40}$.

	Cat	Dog	Total
Grade 9	15	25	40
Grade 10	20	20	40
Total	35	45	80

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

1710

QUESTION ID

S55-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE P(Grade 9 | dog) = 25/45 = 5/9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S55-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.4 • Set 55 • FORWARD 2 • Q5/5

1710

SET 56 • 7.SP.C.7.a • STATISTICS & PROBABILITY

Develop a uniform probability model and use it to find probabilities

MAIN	Grade 7 main question	**7.SP.C.7.a**
BACKWARD 1	Write equivalent fractions	**4.NF.A.1**
BACKWARD 2	Write a fraction for part of a whole	**3.NF.A.1**
FORWARD 1	Understand a function as a rule that gives exactly one output for e…	**8.F.A.1**
FORWARD 2	Use the addition rule for probability	**HSS-CP.B.7**

SET 56 • 7.SP.C.7.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.a

ANSWER KEY Reduced copy of the question page, with its answer

A number cube has faces numbered 1 to 6.

What is the probability of rolling a number greater than 4?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

1713

QUESTION ID

S56-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/3

GRADING NOTE 2/6 is also correct.

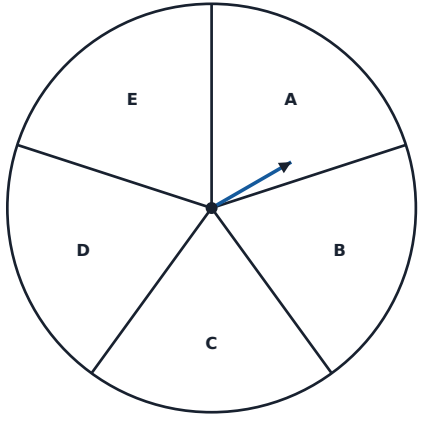
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q1/5

The spinner has 5 equal sections.

What is the probability of landing on a vowel?

Probability: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

1714

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q2/5

A class has 12 girls and 18 boys. One student is chosen at random.

What is the probability that a girl is chosen?

A. $\frac{2}{5}$

B. $\frac{2}{3}$

C. $\frac{1}{12}$

D. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

1715

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q3/5

True or false?

In a uniform probability model, every outcome has the same probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

1716

QUESTION ID

S56-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q4/5

True or false?

A letter is chosen at random from the word MATH.

The probability of choosing a vowel is $\frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

1717

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The probability is 1/4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.a • Set 56 • MAIN • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 1

Write equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Write a fraction equivalent to $\frac{3}{12}$ with a denominator of 4.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

1719

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q1

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q1/5

Write a fraction equivalent to $\frac{10}{15}$ with a denominator of 3.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

1720

QUESTION ID

S56-B1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q2

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{2}{3}$

B. $\frac{6}{4}$

C. $\frac{4}{12}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

1721

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q3/5

True or false?

$\frac{5}{10} = \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

1722

QUESTION ID

S56-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{2}{8} = \frac{1}{3}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q5/5

1723

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B1-Q5

Grade 4 • Number & Operations–Fractions • 4.NF.A.1 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 7.SP.C.7.a

BACKWARD 2

Write a fraction for part of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

A spinner has 8 equal parts. 3 of the parts are red.

What fraction of the spinner is red?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

1725

QUESTION ID

S56-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A bag has 6 equal-size tiles. 1 tile is blue.

What fraction of the tiles is blue?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

1726

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q2/5

A pizza is cut into 4 equal slices. 3 slices are eaten.

What fraction of the pizza is eaten?

A. $\frac{3}{7}$

B. $\frac{4}{3}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

1727

QUESTION ID

S56-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q3/5

True or false?

One of 5 equal parts is $\frac{1}{5}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

1728

QUESTION ID

S56-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Two of 6 equal parts is $\frac{2}{4}$ of the whole.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

1729

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.1 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 7.SP.C.7.a

FORWARD 1

Understand a function as a rule that gives exactly one output for each input • Grade 8 • Functions • 8.F.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Is the set of pairs (1, 3), (2, 5), (3, 7) a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

1731

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. Each input has exactly one output.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q1/5 • NEAREST RELATED

Is the set of pairs (2, 4), (2, 6), (3, 8) a function? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

1732

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. The input 2 has two different outputs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which table shows a function?

A. x: 1, 1, 2 and y: 3, 5, 7

B. x: 1, 2, 3 and y: 4, 4, 4

C. x: 0, 0, 1 and y: 0, 1, 1

D. x: 5, 5, 5 and y: 1, 2, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

1733

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 2, 3 and y: 4, 4, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A probability model gives exactly one probability to each outcome, just as a function gives exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

1734

QUESTION ID

S56-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a function, one input can have two different outputs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

1735

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 56 • 7.SP.C.7.a

FORWARD 2

Use the addition rule for probability • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7

ANSWER KEY Reduced copy of the question page, with its answer

A card is chosen at random from cards numbered 1 to 10.

Find P(even or greater than 7).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

1737

QUESTION ID

S56-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

GRADING NOTE 6/10 or 0.6 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q1/5

P(A) = 0.4, P(B) = 0.5, and P(A and B) = 0.2.

Find P(A or B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

1738

QUESTION ID

S56-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S56-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q2/5

A number cube is rolled.

What is P(less than 3 or odd)?

A. $\frac{5}{6}$

B. $\frac{2}{3}$

C. $\frac{1}{3}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

1739

QUESTION ID

S56-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q3/5

True or false?

If A and B cannot both happen, then P(A or B) = P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

1740

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

P(A or B) is always equal to P(A) + P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

1741

QUESTION ID

S56-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE You subtract P(A and B) when A and B can happen together.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S56-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.B.7 • Set 56 • FORWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b • STATISTICS & PROBABILITY

Develop a probability model from observed frequencies

MAIN	Grade 7 main question	**7.SP.C.7.b**
BACKWARD 1	Read a scaled bar graph	**3.MD.B.3**
BACKWARD 2	Write a part of a whole as a percent	**6.RP.A.3.c**
FORWARD 1	Construct a two-way table from collected data — nearest related…	**8.SP.A.4**
FORWARD 2	Summarize categorical data with relative frequencies	**HSS-ID.B.5**

SET 57 • 7.SP.C.7.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7.b

ANSWER KEY Reduced copy of the question page, with its answer

A bottle cap is tossed 80 times. It lands top-up 52 times.

Estimate the probability that it lands top-up.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

1744

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

0.65

GRADING NOTE 52/80 or 13/20 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A spinner with three unequal sections is spun 200 times: red 90 times, blue 70 times, and green 40 times.

Write an estimated probability for each color.

Probabilities: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

1745

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

Red 0.45, blue 0.35, green 0.2

GRADING NOTE Equivalent fractions or percents are also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q2/5

A spinner landed on red 90 times, blue 70 times, and green 40 times in 200 spins.

Which is the best prediction for 1,000 spins?

A. About 900 reds

B. About 333 reds

C. About 90 reds

D. About 450 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q3/5

1746

QUESTION ID

S57-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. About 450 reds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q3/5

True or false?

A probability estimated from observed frequencies can change when more trials are done.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

1747

QUESTION ID

S57-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q4/5

True or false?

A model built from 10 tosses is just as reliable as a model built from 1,000 tosses.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

1748

QUESTION ID

S57-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7.b • Set 57 • MAIN • Q5/5

SET 57 • 7.SP.C.7.b

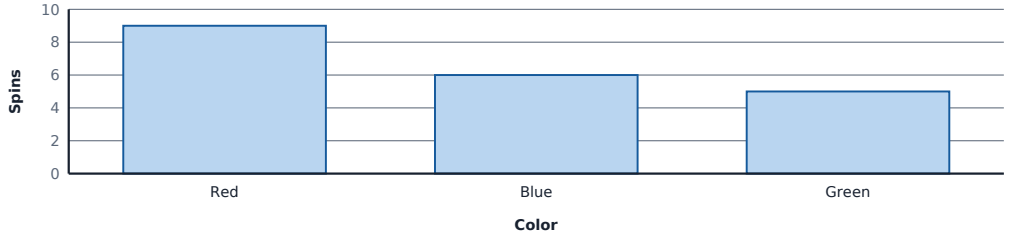
BACKWARD 1

Read a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many spins landed on red?



Spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

1750

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q1

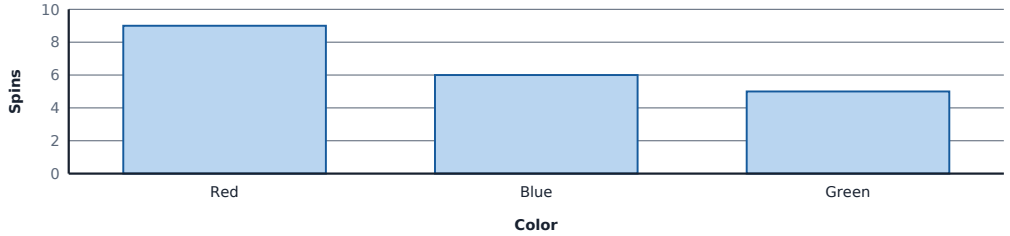
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q1/5

1750

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many more spins landed on red than on green?



Spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

1751

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

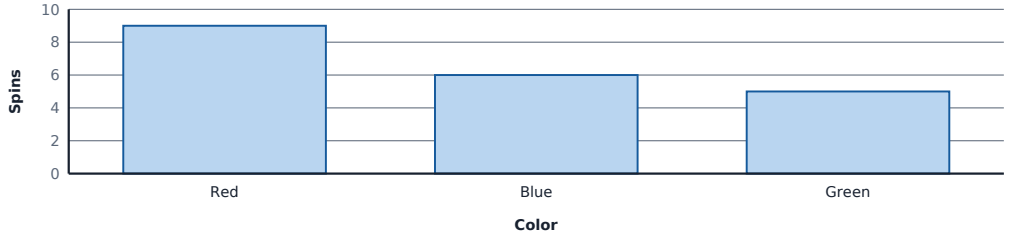
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

The bar graph shows the results of some spins.

How many spins are shown in all?



A. 9 **B.** 11 **C.** 15 **D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

1752

QUESTION ID

S57-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

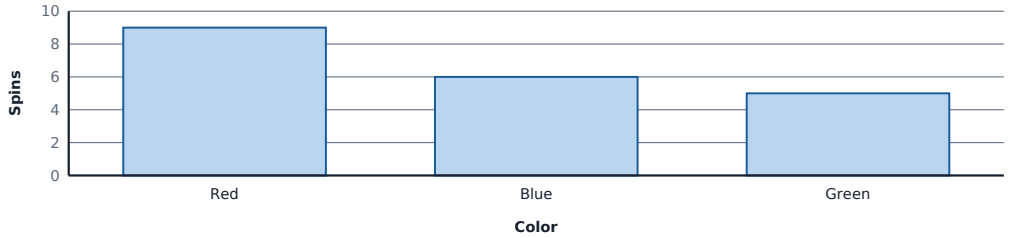
D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q3/5

True or false?

The bar graph shows that blue was spun 6 times.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

1753

QUESTION ID

S57-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

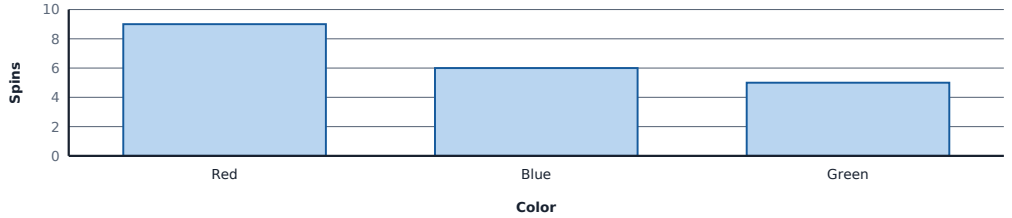
ID S57-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The bar graph shows that green was spun more times than blue.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

1754

QUESTION ID

S57-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 57 • BACKWARD 1 • Q5/5

SET 57 • 7.SP.C.7.b

BACKWARD 2

Write a part of a whole as a percent • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

ANSWER KEY Reduced copy of the question page, with its answer

Write 45 out of 60 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

1756

QUESTION ID

S57-B2-Q1

TYPE

Short answer

CORRECT ANSWER

75%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q1/5

Write 13 out of 50 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

1757

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

26%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q2/5

What percent is 72 out of 120?

A. 72%

B. 48%

C. 60%

D. 40%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

1758

QUESTION ID

S57-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 60%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q3/5

True or false?

9 out of 36 is 25%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

1759

QUESTION ID

S57-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

14 out of 40 is 14%.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

1760

QUESTION ID

S57-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 14 out of 40 is 35%.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 7.SP.C.7.b

FORWARD 1

Construct a two-way table from collected data • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

How many students have neither a phone nor a tablet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

1762

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

50 students were surveyed. 30 have a phone. 18 of the students with a phone also have a tablet. 8 of the students without a phone have a tablet.

How many students have a tablet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

1763

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q2/5 • NEAREST RELATED

A two-way table has the rows "Phone" and "No phone" and the columns "Tablet" and "No tablet."

How many cells show counts for both variables (not totals)?

A. 2

B. 9

C. 6

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

1764

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A two-way table shows how often each combination of two categories happens.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

1765

QUESTION ID

S57-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

In a two-way table, the row totals and the column totals add to different grand totals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

1766

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 57 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 57 • 7.SP.C.7.b

FORWARD 2

Summarize categorical data with relative frequencies • Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the joint relative frequency of Grade 9 students who play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

1768

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.18

GRADING NOTE 18/100 or 18% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q1

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

The table shows the results of a survey.

What is the marginal relative frequency of students who do not play music?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q2

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

1769

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

0.7

GRADING NOTE 70/100 or 70% is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S57-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q2/5

The table shows the results of a survey.

What is the conditional relative frequency of playing music among Grade 10 students?

	Plays music	No music	Total
Grade 9	18	42	60
Grade 10	12	28	40
Total	30	70	100

A. 0.7 **B.** 0.4 **C.** 0.3 **D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

1770

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q3

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q3/5

True or false?

A marginal relative frequency uses a row total or a column total divided by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

1771

QUESTION ID

S57-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q4

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q4/5

True or false?

A conditional relative frequency always divides by the grand total.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

1772

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S57-F2-Q5

Grade 9 (High School) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5 • Set 57 • FORWARD 2 • Q5/5

SET 58 • 7.SP.C.7 • STATISTICS & PROBABILITY

Compare probabilities from a model with observed frequencies and explain any differences

MAIN	Grade 7 main question	**7.SP.C.7**
BACKWARD 1	Find a fraction of a whole number	**5.NF.B.4.a**
BACKWARD 2	Divide whole numbers to find equal shares	**4.NBT.B.6**
FORWARD 1	Compare a linear model's predictions with observed data — near…	**8.SP.A.3**
FORWARD 2	Decide whether observed results are consistent with a model	**HSS-IC.A.2**

SET 58 • 7.SP.C.7

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.7

ANSWER KEY Reduced copy of the question page, with its answer

A fair number cube is rolled 60 times.

How many times would you expect to roll a 5?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

1775

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q1/5

A fair coin is flipped 100 times and lands heads 61 times.

Compare this with the model. Give a possible reason for the difference.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

1776

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

The model predicts about 50 heads. Random variation can explain some of the difference, but 61 heads might also mean the coin is not fair.

GRADING NOTE Any answer that compares with the expected 50 and gives a reasonable cause (chance variation or an unfair coin).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q2/5

A spinner has 4 equal sections, one of them yellow. In 80 spins, it lands on yellow 35 times.

Which conclusion is best?

A. The model predicts 40 yellows.

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

C. Yellow should come up 35 times in every 80 spins.

D. The results match the model exactly.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

1777

QUESTION ID

S58-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Yellow came up much more than the expected 20 times, so the spinner may not be fair.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q3/5

True or false?

A good probability model does not have to match observed frequencies exactly.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

1778

QUESTION ID

S58-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q4/5

True or false?

A coin that lands heads 7 times in 10 flips must be unfair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

1779

QUESTION ID

S58-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.7 • Set 58 • MAIN • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 1

Find a fraction of a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Find the value.

$\frac{1}{6} \times 60$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

1781

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

What is $\frac{1}{4}$ of 80?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

1782

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q2/5

What is $\frac{2}{5}$ of 50?

A. 25

B. 125

C. 20

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

1783

QUESTION ID

S58-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3}$ of 90 is 30.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

1784

QUESTION ID

S58-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{3}{4}$ of 40 is 10.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

1785

QUESTION ID

S58-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 3/4 of 40 is 30.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 7.SP.C.7

BACKWARD 2

Divide whole numbers to find equal shares • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

ANSWER KEY Reduced copy of the question page, with its answer

Divide.

240 ÷ 4

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

1787

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q1/5

Divide.

300 ÷ 6

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

1788

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q2/5

Divide.

1,000 ÷ 8

A. 105

B. 150

C. 125

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

1789

QUESTION ID

S58-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q3/5

True or false?

500 ÷ 5 = 100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

1790

QUESTION ID

S58-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q4/5

True or false?

720 ÷ 3 = 24

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

1791

QUESTION ID

S58-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 7.SP.C.7

FORWARD 1

Compare a linear model's predictions with observed data • Grade 8 • Statistics & Probability • 8.SP.A.3

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

A model is y = 2x + 1. At x = 4, the observed value is y = 11.

How far is the observed value from the prediction?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

1793

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

GRADING NOTE The model predicts 9.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q1/5 • NEAREST RELATED

A model is y = 5x. At x = 3, the observed value is y = 13.

Is the model's prediction too high or too low, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

1794

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Too high, by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q2/5 • NEAREST RELATED

Which observed point is closest to the model y = x + 3?

A. (2, 7)

B. (2, 9)

C. (2, 3)

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

1795

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 5.2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A linear model can be useful even when the data points do not all lie on the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

1796

QUESTION ID

S58-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

If one data point is far from the line of fit, the model must be thrown out.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

1797

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 58 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 58 • 7.SP.C.7

FORWARD 2

Decide whether observed results are consistent with a model • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A spinner is claimed to land on red 25% of the time. In 200 spins it lands on red 52 times.

Is this consistent with the claim? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

1799

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. 52 is close to the expected 50.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q1/5

A number cube is claimed to be fair. In 300 rolls, 6 comes up 120 times.

Is this consistent with the claim? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

1800

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

No. The expected number is about 50, and 120 is far more than that.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S58-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q2/5

A model says P(win) = 0.1. Which number of wins in 50 games would cast the most doubt on the model?

A. 5

B. 4

C. 6

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

1801

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q3/5

True or false?

A simulation can show the range of results that a model usually produces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

1802

QUESTION ID

S58-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q4/5

True or false?

Any difference between observed and expected results proves that a model is wrong.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

1803

QUESTION ID

S58-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S58-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 58 • FORWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a • STATISTICS & PROBABILITY

Find the probability of a compound event as the fraction of outcomes in the sample space

MAIN	Grade 7 main question	**7.SP.C.8.a**
BACKWARD 1	Multiply fractions	**5.NF.B.4.a**
BACKWARD 2	Use multiplication to count combinations in equal groups	**3.OA.A.3**
FORWARD 1	Find joint relative frequencies in a two-way table — nearest relate…	**8.SP.A.4**
FORWARD 2	Use the multiplication rule for independent events	**HSS-CP.A.2**

SET 59 • 7.SP.C.8.a

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A coin is flipped and a number cube is rolled.

What is the probability of getting heads and a 6?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

1806

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two coins are flipped.

What is the probability that both land tails?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

1807

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q2/5

The spinner has 3 equal sections. It is spun twice.

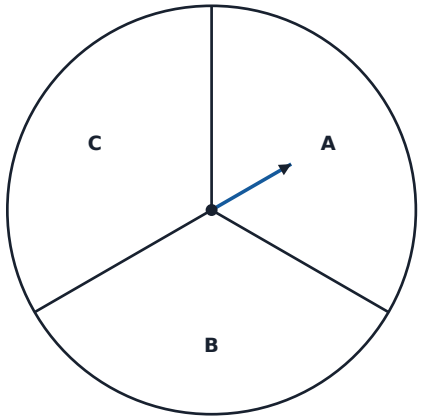
What is the probability of landing on A both times?

A. $\frac{2}{3}$

B. $\frac{1}{6}$

C. $\frac{1}{9}$

D. $\frac{1}{3}$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

1808

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q3/5

True or false?

Two number cubes are rolled. The probability that the sum is 7 is $\frac{6}{36} = \frac{1}{6}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

1809

QUESTION ID

S59-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

Two coins are flipped. The probability of getting one head and one tail is $\frac{1}{4}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

1810

QUESTION ID

S59-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The probability is 2/4 = 1/2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.a • Set 59 • MAIN • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 1

Multiply fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{2} \times \frac{1}{6}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

1812

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{2}{3} \times \frac{3}{4}$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

1813

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/2

GRADING NOTE 6/12 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{4} \times \frac{1}{4}$

A. $\frac{2}{4}$

B. $\frac{1}{8}$

C. $\frac{1}{16}$

D. $\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

1814

QUESTION ID

S59-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{3} \times \frac{1}{2} = \frac{1}{6}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

1815

QUESTION ID

S59-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{5} \times \frac{1}{5} = \frac{2}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

1816

QUESTION ID

S59-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B1-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 59 • BACKWARD 1 • Q5/5

SET 59 • 7.SP.C.8.a

BACKWARD 2

Use multiplication to count combinations in equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

A café has 3 kinds of sandwiches and 4 kinds of drinks.

How many different sandwich-and-drink pairs can be made?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

1818

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q1/5

Ty has 4 shirts and 5 pairs of pants.

How many different shirt-and-pants outfits can he make?

Outfits: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

1819

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q2/5

A shop has 6 ice cream flavors and 2 kinds of cones.

How many different flavor-and-cone choices are there?

A. 12

B. 8

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

1820

QUESTION ID

S59-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q3/5

1820

True or false?

3 rows of 7 chairs is 21 chairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

1821

QUESTION ID

S59-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q4/5

True or false?

5 hats and 3 scarves make 8 different hat-and-scarf pairs.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

1822

QUESTION ID

S59-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make 15 pairs.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 7.SP.C.8.a

FORWARD 1

Find joint relative frequencies in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

ANSWER KEY Reduced copy of the question page, with its answer

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of an even number and heads?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

1824

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

0.24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q1/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

The table shows 100 trials of flipping a coin and rolling a number cube.

What is the relative frequency of tails?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

Relative frequency: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

1825

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q2/5 • NEAREST RELATED

The table shows 100 trials of flipping a coin and rolling a number cube.

Which result happened most often?

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. Even and tails

B. Odd and tails

C. Even and heads

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

1826

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Odd and heads

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

In the table, the relative frequency of an odd number and tails is 0.23.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

1827

QUESTION ID

S59-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q4/5 • NEAREST RELATED

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The relative frequencies of the four results in the table add to 2.

	Heads	Tails	Total
Even	24	26	50
Odd	27	23	50
Total	51	49	100

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

1828

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They add to 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 59 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 59 • 7.SP.C.8.a

FORWARD 2

Use the multiplication rule for independent events • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2

Events A and B are independent. P(A) = 0.3 and P(B) = 0.5.

Find P(A and B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

1830

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q1**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q1/5

A fair coin is flipped 3 times.

What is the probability that all 3 flips are heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

1831

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q2/5

A fair number cube is rolled twice.

What is the probability of rolling two 6s?

A. $\frac{1}{6}$

B. $\frac{1}{36}$

C. $\frac{1}{12}$

D. $\frac{2}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

1832

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 1/36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S59-F2-Q3**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q3/5

True or false?

Events A and B are independent when P(A and B) = P(A) × P(B).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q4/5

1833

QUESTION ID

S59-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q4/5

True or false?

Drawing two cards from a deck without putting the first card back are independent events.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

1834

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S59-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.2 • Set 59 • FORWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b • STATISTICS & PROBABILITY

Represent the sample space of a compound event with a list, a table, or a tree diagram

MAIN	Grade 7 main question	**7.SP.C.8.b**
BACKWARD 1	Use repeated addition to find the total in an array	**2.OA.C.4**
BACKWARD 2	Form ordered pairs from two numerical patterns	**5.OA.B.3**
FORWARD 1	Organize the outcomes of two categorical variables in a two-way…	**8.SP.A.4**
FORWARD 2	Find a conditional probability as P(A and B) ÷ P(B)	**HSS-CP.A.3**

SET 60 • 7.SP.C.8.b

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.b

A coin is flipped and a spinner with sections 1, 2, and 3 is spun.

List the sample space.

Sample space: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

1837

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

H1, H2, H3, T1, T2, T3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

A café has 3 sandwiches and 4 drinks.

How many outcomes are in the sample space for choosing one sandwich and one drink?

Outcomes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

1838

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q2/5

A tree diagram shows choosing a shirt (red or blue) and then pants (black, tan, or gray).

How many branches are at the end of the tree?

A. 3

B. 2

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

1839

QUESTION ID

S60-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q3/5

True or false?

"Roll a 3 and get tails" is one outcome in the sample space for rolling a number cube and flipping a coin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

1840

QUESTION ID

S60-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q4/5

True or false?

The sample space for flipping two coins is HH, HT, TT.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

1841

QUESTION ID

S60-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE TH is missing.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.b • Set 60 • MAIN • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 1

Use repeated addition to find the total in an array • Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4

An array has 3 rows with 4 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

1843

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 + 4 + 4 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q1/5

An array has 2 rows with 5 objects in each row.

Write an addition equation for the total.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

1844

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 + 5 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q2/5

Which equation gives the total of an array with 4 rows of 3?

A. 4 + 3 = 7

B. 3 + 3 + 3 + 3 = 12

C. 3 + 3 + 3 = 9

D. 3 + 4 + 3 = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

1845

QUESTION ID

S60-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 + 3 + 3 + 3 = 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q3/5

True or false?

An array with 3 rows of 5 has 5 + 5 + 5 = 15 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

1846

QUESTION ID

S60-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q4/5

True or false?

An array with 2 rows of 4 has 4 + 2 = 6 objects.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

1847

QUESTION ID

S60-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B1-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.C.4 • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 7.SP.C.8.b

BACKWARD 2

Form ordered pairs from two numerical patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A starts at 0 and adds 2. Pattern B starts at 0 and adds 4.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

1849

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(0, 0), (2, 4), (4, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Pattern A starts at 1 and adds 1. Pattern B starts at 3 and adds 3.

Write the first 3 ordered pairs (A, B).

Ordered pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

1850

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(1, 3), (2, 6), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q2/5

Pattern A is 0, 3, 6, 9. Pattern B is 0, 6, 12, 18.

Which ordered pair (A, B) comes from these patterns?

A. (3, 3)

B. (12, 6)

C. (6, 9)

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

1851

QUESTION ID

S60-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q3/5

True or false?

In the patterns A: 0, 5, 10 and B: 0, 10, 20, each term in B is twice the matching term in A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

1852

QUESTION ID

S60-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q4/5

True or false?

The patterns A: 0, 2, 4 and B: 0, 3, 6 make the ordered pair (4, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

1853

QUESTION ID

S60-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They make the pair (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 60 • BACKWARD 2 • Q5/5

SET 60 • 7.SP.C.8.b

FORWARD 1

Organize the outcomes of two categorical variables in a two-way table • Grade 8 • Statistics & Probability • 8.SP.A.4

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

How many students have a sibling but no pet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q1/5 • NEAREST RELATED

1855

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q1/5 • NEAREST RELATED

In a class, 14 students have a sibling and 10 do not. 8 of the students with a sibling have a pet. 3 of the students without a sibling have a pet.

How many students have a pet?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

1856

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q2/5 • NEAREST RELATED

A two-way table has the rows "Sibling" and "No sibling" and the columns "Pet" and "No pet."

How many outcome cells (not totals) does it have?

A. 4

B. 2

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

1857

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A two-way table can show the counts for two categorical variables at the same time.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

1858

QUESTION ID

S60-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

A two-way table with 3 rows and 2 columns of outcomes has 5 outcome cells.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

1859

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 3 × 2 = 6 outcome cells.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 60 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 60 • 7.SP.C.8.b

FORWARD 2

Find a conditional probability as P(A and B) ÷ P(B) • Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3

ANSWER KEY Reduced copy of the question page, with its answer

P(A and B) = 0.12 and P(B) = 0.4.

Find P(A | B).

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

1861

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q1

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q1/5

Two coins are flipped. At least one landed heads.

What is the probability that both landed heads?

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q2

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

1862

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S60-F2-Q2**

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q2/5

A number cube is rolled, and the result is even.

What is the probability that it is a 6?

A. $\frac{1}{6}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. $\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q3/5

1863

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q3

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q3/5

True or false?

If A and B are independent, then P(A | B) = P(A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

1864

QUESTION ID

S60-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q4

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q4/5

True or false?

P(A | B) is always equal to P(B | A).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q5/5

1865

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S60-F2-Q5

Grade 9 (High School) • Statistics & Probability: Conditional Probability & the Rules of Probability • HSS-CP.A.3 • Set 60 • FORWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c • STATISTICS & PROBABILITY

Design and use a simulation to estimate the probability of a compound event

MAIN	Grade 7 main question	**7.SP.C.8.c**
BACKWARD 1	Display data in a dot plot	**6.SP.B.4**
BACKWARD 2	Compare fractions with the same numerator or the same denomin…	**3.NF.A.3.d**
FORWARD 1	Construct and interpret a scatter plot — nearest related (no direct…	**8.SP.A.1**
FORWARD 2	Use a simulation to decide whether a difference between treatme…	**HSS-IC.B.5**

SET 61 • 7.SP.C.8.c

MAIN

Grade 7 main question • Grade 7 • Statistics & Probability • 7.SP.C.8.c

ANSWER KEY Reduced copy of the question page, with its answer

A player makes 60% of her free throws.

Describe how to use random digits 0 to 9 to simulate one trial of 2 free throws.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

1868

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

Let the digits 0 to 5 stand for a make and 6 to 9 stand for a miss. Choose 2 random digits; each pair of digits is one trial.

GRADING NOTE Any plan where 6 of the 10 digits stand for a make, and each trial uses 2 digits, is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q1

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In a simulation with 50 trials, 18 trials showed both free throws made.

Estimate the probability that both are made.

Probability: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

1869

QUESTION ID
S61-M-Q2

TYPE
Short answer

CORRECT ANSWER
0.36

GRADING NOTE 18/50 or 9/25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q2

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q2/5

Each of 3 children is equally likely to be a boy or a girl.

Which tool can simulate the 3 children?

A. Spin a spinner with 3 unequal parts once

B. Flip a coin 3 times

C. Draw one card from a deck

D. Roll one number cube once

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q3/5

1870

QUESTION ID

S61-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Flip a coin 3 times

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q3

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q3/5

True or false?

Running more trials of a simulation usually gives a better estimate of a probability.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

1871

QUESTION ID

S61-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q4/5

True or false?

A single spin of a spinner with 4 equal sections can simulate an event with a probability of $\frac{1}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

1872

QUESTION ID

S61-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Four equal sections give probabilities in fourths; use 3 equal sections instead.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-M-Q5

Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 61 • MAIN • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 1

Display data in a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

Make a dot plot of the numbers of heads in 8 trials:

1, 2, 2, 3, 1, 2, 0, 2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

1874

QUESTION ID

S61-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Dots above 0: 1, above 1: 2, above 2: 4, above 3: 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q1/5

The dot plot shows the number of heads in 10 trials.

In how many trials were there 2 or more heads?



Trials: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

1875

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q2/5

Which display shows each data value as a dot above a number line?

A. Dot plot

B. Histogram

C. Bar graph

D. Box plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

1876

QUESTION ID

S61-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Dot plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q3/5

True or false?

A dot plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

1877

QUESTION ID

S61-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A box plot shows every value in a data set.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

1878

QUESTION ID

S61-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 7.SP.C.8.c

BACKWARD 2

Compare fractions with the same numerator or the same denominator • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d

ANSWER KEY Reduced copy of the question page, with its answer

Which is greater, $\frac{3}{8}$ or $\frac{5}{8}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

1880

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q1

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Write >, <, or =.

$\frac{2}{3}$ ___ $\frac{2}{6}$

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

1881

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

>

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q2

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q2/5

Which fraction is less than $\frac{4}{6}$?

A. $\frac{5}{6}$

B. $\frac{6}{6}$

C. $\frac{3}{6}$

D. $\frac{4}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

1882

QUESTION ID

S61-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q3

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{4} < \frac{1}{2}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

1883

QUESTION ID

S61-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q4

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$\frac{3}{5} > \frac{4}{5}$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

1884

QUESTION ID

S61-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-B2-Q5

Grade 3 • Number & Operations–Fractions • 3.NF.A.3.d • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 7.SP.C.8.c

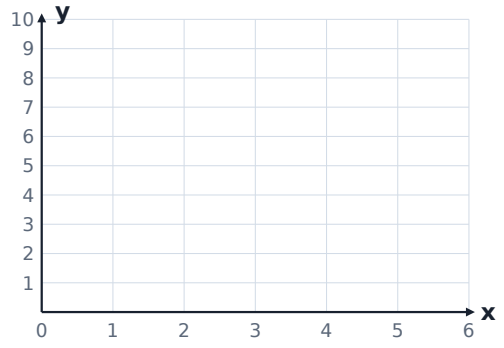
FORWARD 1

Construct and interpret a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

⚑ No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data in the table.

x	1	2	3	4	5
y	2	4	5	7	9



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

1886

QUESTION ID

S61-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (1, 2), (2, 4), (3, 5), (4, 7), and (5, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q1/5 • NEAREST RELATED

The table shows x-values 1, 2, 3, 4, 5 and y-values 2, 4, 5, 7, 9.

Describe the association between x and y.

Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

1887

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A positive linear association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q2/5 • NEAREST RELATED

What does each point in a scatter plot stand for?

A. An outlier

B. One pair of values from the data

C. The mean of the data

D. The line of fit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

1888

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. One pair of values from the data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q3/5 • NEAREST RELATED

True or false?

A scatter plot shows the relationship between two numerical variables.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

1889

QUESTION ID

S61-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F1-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q4/5 • NEAREST RELATED

True or false?

All the points in a scatter plot must lie on a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

1890

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 61 • FORWARD 1 • Q5/5 • NEAREST RELATED

SET 61 • 7.SP.C.8.c

FORWARD 2

Use a simulation to decide whether a difference between treatments is significant • Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

Plants in Group A grew a mean of 12 cm and plants in Group B grew a mean of 9 cm. In 100 re-randomizations of the data, a difference of 3 cm or more happened 2 times.

Is the difference likely due to the treatment? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

1892

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. A difference that large happened by chance only about 2% of the time.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q1

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q1/5

In 100 re-randomizations of the data, a difference of 3 cm or more happened 40 times.

What does this suggest?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

1893

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

The difference could easily be due to chance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q2

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q2/5

Why are the data re-randomized many times in the simulation?

A. To remove outliers

B. To see what differences happen by chance alone

C. To make the mean larger

D. To make the treatment work better

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

1894

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. To see what differences happen by chance alone

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q3

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q3/5

True or false?

Random assignment helps make two treatment groups alike before an experiment begins.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

1895

QUESTION ID

S61-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q4

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q4/5

True or false?

A difference between two groups in one experiment is always caused by the treatment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5

1896

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S61-F2-Q5

Grade 9 (High School) • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 61 • FORWARD 2 • Q5/5